

General Relativity: From Local Conservation to Einstein's Field Equations

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Chapter 1

From Local Conservation to Einstein's Field Equations

Einstein's field equations are simple enough to write in one line and notoriously unpleasant to expand in components. A metric has many Christoffel symbols; the Christoffel symbols contain derivatives of the metric; and the curvature tensor contains both derivatives and products of Christoffel symbols. Even before one tries to solve anything, the calculation can occupy pages. For a particular metric, symbolic software is often the sensible way to manage that bookkeeping.

The principles are much cleaner than the computation. We will therefore not begin by attacking the equations component by component. We will ask what kind of equation gravity requires. The argument starts with Newton's theory, passes through local conservation, and ends by discovering the one combination of curvature tensors that can consistently equal the flow and density of energy and momentum. Throughout the chapter we use $c = 1$.

1.1 The Newtonian Two-Way Street

Gravity has two complementary instructions. The gravitational field tells matter how to move, while matter tells the gravitational field what to be. This two-way street is already visible in Newtonian mechanics.

Let $\phi(\mathbf{x}, t)$ be the Newtonian gravitational potential. A test body of mass m feels the force

$$\mathbf{F} = -m\nabla\phi. \quad (1.1)$$

Combining this with $\mathbf{F} = m\mathbf{a}$ cancels the test mass:

$$\mathbf{a} = -\nabla\phi. \quad (1.2)$$

This is the first direction of the street. Once the field is known, it tells every freely falling body how to accelerate, independently of that body's mass.

The reverse direction is Poisson's equation,

$$\nabla^2\phi = 4\pi G\rho, \quad \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}, \quad (1.3)$$

where $\rho(\mathbf{x}, t)$ is mass per unit volume. The factor 4π could be moved into the definition of G , but it would only reappear elsewhere. Equation (1.3) says that the distribution of matter is the source of the potential.

For a spherically symmetric body of total mass M , the exterior solution is

$$\phi(r) = -\frac{GM}{r}. \quad (1.4)$$

The density may be concentrated near the center, spread uniformly, or tapered smoothly toward the surface. Spherical symmetry and the total mass are all that the exterior solution remembers. Outside the matter, $\nabla^2\phi = 0$; differentiating $1/r$ then gives the inverse-square force. If we care only about the exterior, we may shrink the source to a point and use the same expression everywhere except at that point.

Question from the audience. Must the density be constant inside the spherical body and then jump discontinuously to zero at its boundary?

Response. No. It must be spherically symmetric for this argument, but it need not be constant and it need not jump. It may decrease smoothly to zero. None of those interior details changes the exterior potential, which depends only on the total mass. ^a

^aLecture timestamp: 00:09:13–00:09:37.

This field formulation is more local-looking than a direct force between distant masses, but Newtonian gravity still contains instantaneous action at a distance. Move the source, and the Newtonian potential changes everywhere at once. General relativity must replace that instantaneous response as well as the force law.

1.2 A Weak-Field Clue, Not Yet an Equation

The relativistic version of the first instruction is already known: once the spacetime metric is given, freely falling particles follow spacetime geodesics. We now need the relativistic replacement for Poisson's equation, the instruction by which matter determines the metric.

The Schwarzschild geometry offers a clue. Its time-time component is

$$g_{00} = 1 - \frac{2GM}{r}. \quad (1.5)$$

Using Equation (1.4), this becomes

$$g_{00} = 1 + 2\phi. \quad (1.6)$$

Consequently,

$$\nabla^2 g_{00} = 2\nabla^2\phi = 8\pi G\rho. \quad (1.7)$$

Equation (1.7) must be taken with a grain of salt. It is a mnemonic, not Einstein's equation. It uses a preferred component of the metric and a nonrelativistic source. Still, it displays the texture we are looking for: a matter density on one side and second derivatives of geometry on the other. It is the rough bridge from Poisson's equation to a covariant field equation.

The right-hand side now needs closer attention. In relativistic units mass is energy, so ρ should first be understood as an energy density. But an energy density is not a complete relativistic object. To see what accompanies it, we begin with a simpler conserved quantity.

1.3 Charge Density, Current, and Local Conservation

Let Q be electric charge and let σ denote charge density:

$$\sigma = \lim_{\Delta V \rightarrow 0} \frac{\Delta Q}{\Delta V}. \quad (1.8)$$

Density asks how much charge occupies a small spatial volume at one instant. Current asks a different question. Place an infinitesimal oriented window in space and count how much charge crosses it per unit area per unit time. The three possible orientations give the spatial current components J^i .

The dimensional comparison is instructive. Charge density is charge divided by three spatial lengths. Current is charge divided by two spatial lengths and one time interval. When $c = 1$, time and length have the same units, so density and current have matching dimensions. They fit together as a four-vector:

$$J^\mu = (J^0, J^1, J^2, J^3), \quad J^0 = \sigma. \quad (1.9)$$

Charge conservation is not merely a statement that the total charge in the universe stays fixed. It is local. Charge cannot vanish from the classroom and instantaneously reappear at Alpha Centauri. If it leaves a region, it must cross the region's boundary. For a small box, the decrease of charge inside equals the outward flux through its walls. Gauss's theorem gives

$$\dot{\sigma} + \nabla \cdot \mathbf{J} = 0, \quad (1.10)$$

or, using the summation convention,

$$\partial_\mu J^\mu = 0. \quad (1.11)$$

In curved coordinates the corresponding tensor equation is written with a covariant derivative:

$$D_\mu J^\mu = 0. \quad (1.12)$$

The purpose of the covariant derivative is not decoration. It makes the statement meaningful in any coordinate system.

1.4 Why Energy and Momentum Need Two Indices

Charge itself is invariant. Observers in relative motion agree on the charge of an electron, although they do not agree on the charge density or current. A moving observer sees Lorentz-contracted volumes and may see a wind of charges where another observer sees charges at rest. Density and current therefore mix under a change of frame even though total charge does not.

Energy and momentum are more complicated. Even their totals are frame-dependent. One observer may see a particle at rest, carrying only its rest energy and no spatial momentum. Another observer walking past sees both kinetic energy and momentum. The four quantities form the momentum four-vector

$$P^\nu = (P^0, P^1, P^2, P^3), \quad P^0 = E. \quad (1.13)$$

Each component P^ν is locally conserved. Each therefore has both a density and a flow. One index identifies the conserved quantity; a second index identifies density or direction of flow.

Start with energy. Its density is T^{00} . Energy can also cross a window, so T^{0i} is the flow of energy in the x^i direction. Now take the i -component of momentum. Its density is T^{i0} , and its flow in the x^j direction is T^{ij} . In this convention:

- T^{00} is energy density;
- T^{0i} is energy flux in direction i ;
- T^{i0} is density of the i -component of momentum;
- T^{ij} is flux of i -momentum through a surface normal to j .

The spatial block T^{ij} will later be recognized as stress: its diagonal entries include pressures, while its off-diagonal entries describe shear.

Question from the audience. When $T^{m0}(x)$ is called the density of the m -component of momentum, does x itself mean the m -component of position?

Response. No. The argument x denotes the spacetime position at which the density is evaluated; it stands for all the coordinates. The index m labels which momentum component is being counted. ^a

^aLecture timestamp: 00:41:07–00:41:27.

For each fixed ν , local conservation gives a continuity equation:

$$D_\mu T^{\nu\mu} = 0. \quad (1.14)$$

The four values of ν give one equation for energy and three for momentum. Together the densities and flows form the energy-momentum tensor.

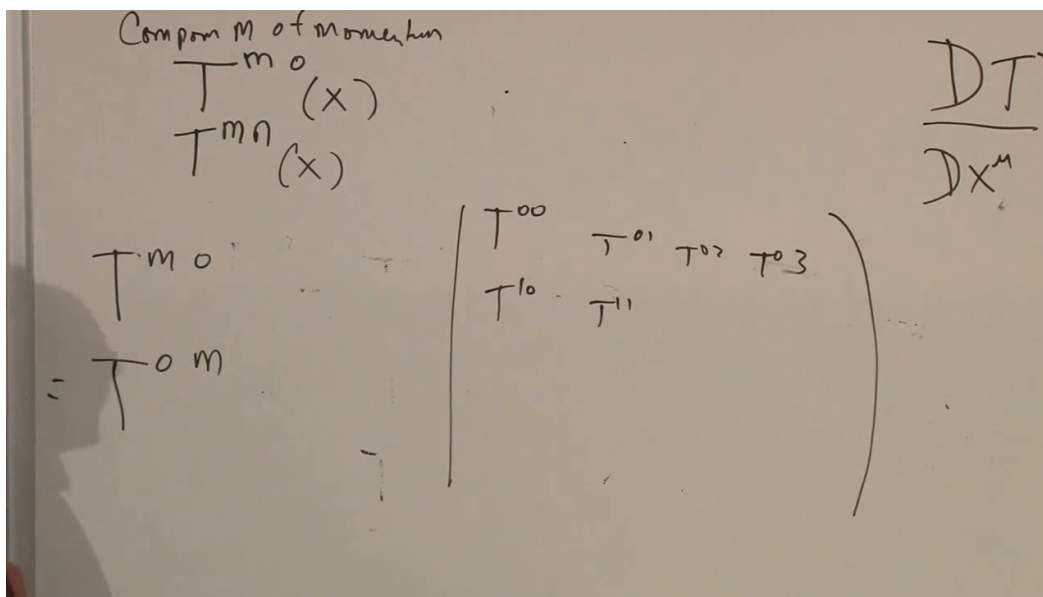


Figure 1.1: The developing energy-momentum array on the board. The first index selects energy or a momentum component; the second selects density or flow.

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The full array is

$$T^{\mu\nu} = \begin{pmatrix} T^{00} & T^{01} & T^{02} & T^{03} \\ T^{10} & T^{11} & T^{12} & T^{13} \\ T^{20} & T^{21} & T^{22} & T^{23} \\ T^{30} & T^{31} & T^{32} & T^{33} \end{pmatrix}. \quad (1.15)$$

A further theorem of relativistic mechanics and field theory makes the stress-energy tensor relevant to gravity symmetric:

$$T^{\mu\nu} = T^{\nu\mu}. \quad (1.16)$$

In particular, energy flux and momentum density are the corresponding off-diagonal entries viewed from opposite sides. Symmetry also lets us write Equation (1.14) in the familiar form

$$D_\mu T^{\mu\nu} = 0. \quad (1.17)$$

Question from the audience. Why did energy-momentum acquire an extra index when charge current needed only one?

Response. The current index already distinguishes density from flow direction. Charge needs no additional label because the conserved charge is a scalar. Energy and the three momenta are themselves four components, so the first index says which conserved quantity is involved and the second says whether it is a density or a flow. ^a

^aLecture timestamp: 00:45:31–00:47:24.

Question from the audience. Is the entire $T^{\mu\nu}$ matrix symmetric, not merely the energy-flux and momentum-density entries?

Response. Yes. The entire matrix is symmetric. The detailed meaning of its spatial entries, including pressure and stress, will be developed later. ^a

^aLecture timestamp: 00:47:32–00:47:54.

Energy density was therefore only one corner of a larger object. The source of gravity cannot consistently be T^{00} alone; it must involve the full tensor.

1.5 A Law Cannot Select One Component

The need for a tensor equation can be seen without any curvature calculation. Suppose A_i and B_i are vectors and a proposed law says only $A_3 = B_3$. If that equality is to hold in every rotated frame, rotate the third axis into the second. The same law then says $A_2 = B_2$; another rotation gives $A_1 = B_1$. A frame-independent law was really the vector equation

$$A_i = B_i. \quad (1.18)$$

Lorentz transformations extend the same argument to time. If $A^0 = B^0$ is to hold in every inertial frame, boosts mix the time component with spatial components and force the full equation $A^\mu = B^\mu$. Good relativistic laws must equate tensors of the same type.

The Newtonian mnemonic singled out T^{00} , but the relativistic equation must have the form

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}, \quad (1.19)$$

where the symbol $G_{\mu\nu}$ is initially only a placeholder. Its required properties are now clear:

1. it must be a symmetric rank-two tensor;
2. it must be made from the metric, not from matter;
3. it should contain two derivatives of the metric;
4. its covariant divergence must vanish.

The last condition follows from Equation (1.17).

1.6 Searching Among Curvature Tensors

The Christoffel symbols contain one derivative of the metric:

$$\Gamma_{\tau\nu}^{\sigma} = \frac{1}{2}g^{\sigma\delta}(\partial_{\tau}g_{\delta\nu} + \partial_{\nu}g_{\delta\tau} - \partial_{\delta}g_{\tau\nu}). \quad (1.20)$$

They are not tensors, so they cannot by themselves provide the left-hand side. The Riemann curvature tensor is a tensor:

$$R^{\rho}{}_{\sigma\mu\nu} = \partial_{\mu}\Gamma_{\nu\sigma}^{\rho} - \partial_{\nu}\Gamma_{\mu\sigma}^{\rho} + \Gamma_{\mu\lambda}^{\rho}\Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\nu\lambda}^{\rho}\Gamma_{\mu\sigma}^{\lambda}. \quad (1.21)$$

Its first two terms contain second derivatives of the metric; its last two contain products of first derivatives. This is exactly the differential order suggested by the Newtonian equation.

The difficulty is index structure. $R^{\rho}{}_{\sigma\mu\nu}$ has four indices, while $T_{\mu\nu}$ has two. Contracting a suitable upper and lower index gives the Ricci tensor,

$$R_{\mu\nu} = R^{\lambda}{}_{\mu\lambda\nu}. \quad (1.22)$$

Other apparent contractions either vanish because of the symmetries of the Riemann tensor or reproduce the same object up to a sign and index convention. The Ricci tensor is symmetric, but it has less information than the full Riemann tensor: it can vanish while the Riemann tensor remains nonzero.

Question from the audience. Can one simply contract two lower indices of the Riemann tensor? Does not a contraction require one covariant and one contravariant index?

Response. Exactly. An index must first be raised with the inverse metric, or equivalently the contraction must include $g^{\mu\nu}$. Written as $R^{\lambda}{}_{\mu\lambda\nu}$, the required upper-lower pair is explicit. ^a

^aLecture timestamp: 01:00:09–01:00:33.

Contracting the Ricci tensor again gives the curvature scalar,

$$R = g^{\mu\nu}R_{\mu\nu}. \quad (1.23)$$

Multiplying this scalar by $g_{\mu\nu}$ returns another symmetric rank-two tensor. We have therefore found two natural candidates:

$$R_{\mu\nu}, \quad g_{\mu\nu}R. \quad (1.24)$$

The exploratory alternatives are

$$R_{\mu\nu} \stackrel{?}{=} 8\pi G T_{\mu\nu}, \quad (1.25)$$

$$g_{\mu\nu}R \stackrel{?}{=} 8\pi G T_{\mu\nu}. \quad (1.26)$$

Neither is to be accepted merely because its indices fit. Conservation provides the test.

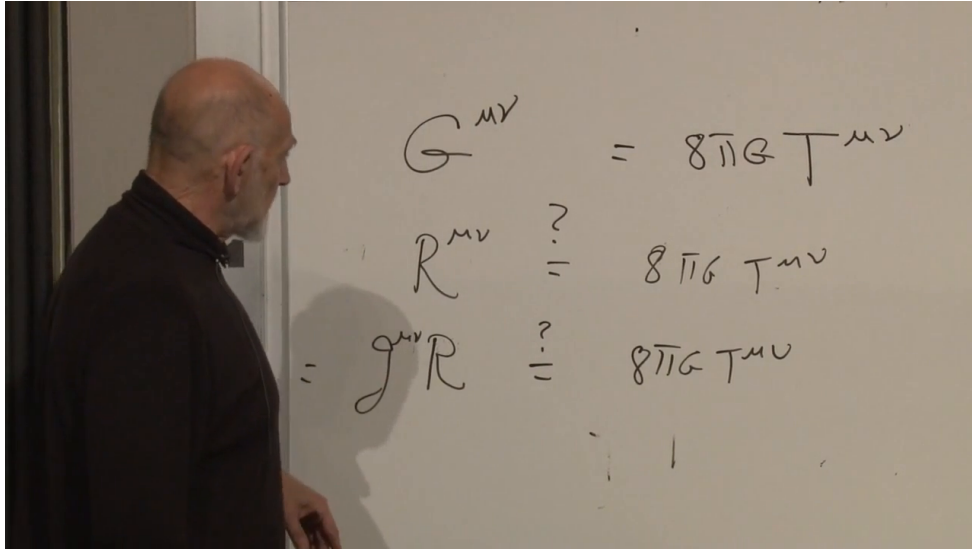


Figure 1.2: The board's two trial geometric tensors. At this stage both have the correct rank and differential order, so each carries a question mark.

Lecture frame: 01:04:56.

1.7 The Divergence Test

If Equation (1.19) is correct, then

$$D_\mu G^{\mu\nu} = 0 \quad (1.27)$$

must follow identically, because $D_\mu T^{\mu\nu} = 0$. Begin with the scalar-curvature candidate. The product rule gives

$$D_\mu (g^{\mu\nu} R) = (D_\mu g^{\mu\nu}) R + g^{\mu\nu} D_\mu R. \quad (1.28)$$

Metric compatibility says

$$D_\mu g^{\alpha\beta} = 0. \quad (1.29)$$

This is an independent geometric property of the Levi-Civita connection, not a consequence of energy-momentum conservation. Since R is a scalar, $D_\mu R = \partial_\mu R$. Equation (1.28) therefore becomes

$$D_\mu (g^{\mu\nu} R) = g^{\mu\nu} \partial_\mu R = D^\nu R, \quad (1.30)$$

which is not generally zero. The candidate $g^{\mu\nu} R$ fails.

The corresponding calculation for the Ricci tensor is somewhat longer. Its result is the contracted Bianchi identity,

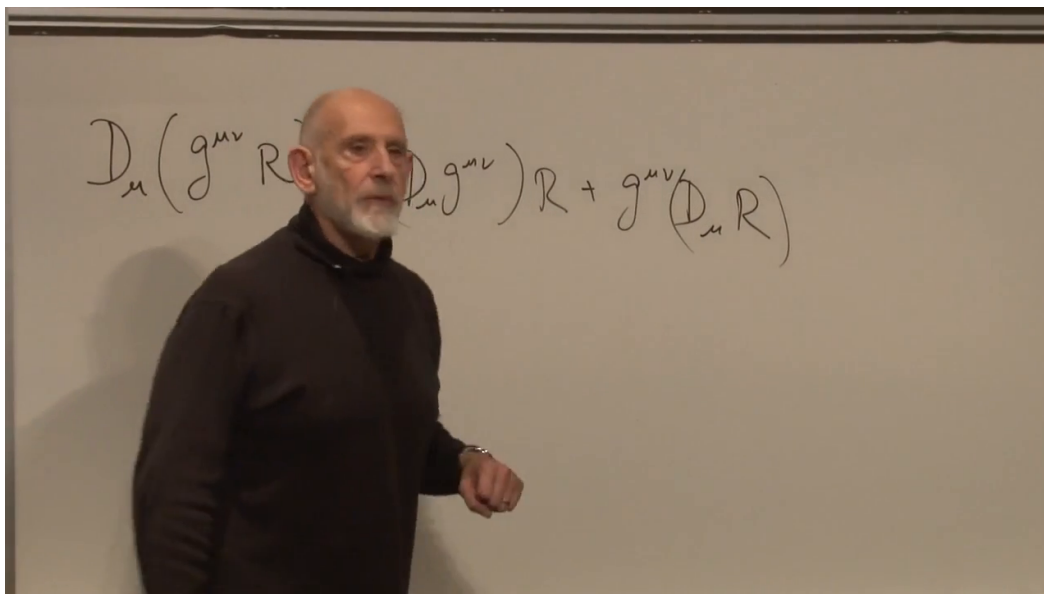
$$D_\mu R^{\mu\nu} = \frac{1}{2} D^\nu R. \quad (1.31)$$

Thus $R^{\mu\nu}$ also fails by itself. But the two failures have precisely the ratio needed for cancellation:

$$D_\mu \left(R^{\mu\nu} - \frac{1}{2} g^{\mu\nu} R \right) = 0. \quad (1.32)$$

This selects the Einstein tensor,

$$\boxed{G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R.} \quad (1.33)$$

Figure 1.3: Applying the covariant product rule to $D_\mu(g^{\mu\nu} R)$.*Lecture frame: 01:07:12.*

Within the assumptions being used here – a local symmetric rank-two tensor made from the metric with no more than two derivatives – this combination is fixed up to an overall coefficient and a possible metric term. The latter is the cosmological-constant possibility, which returns near the end of the lecture. The coefficient multiplying Equation (1.33) is fixed by matching to Newtonian gravity.

Question from the audience. Is there a more directly physical way to obtain the field equation, rather than trying candidates and rejecting them with the continuity equation?

Response. Local conservation is already a physical principle: energy and momentum cannot disappear from a region without crossing its boundary. There is also a more condensed derivation from an action principle for the gravitational field. Varying that action produces the same equation, although the calculation itself is not short. ^a

^aLecture timestamp: 01:17:14–01:18:41.

Question from the audience. Could some other geometrical rank-two tensor satisfy the same conservation law? Is the answer physical or mathematical, and would Einstein have known it?

Response. Under the assumptions of this construction, the classification is mathematical. The available two-derivative curvature objects descend from the Riemann tensor; the two rank-two candidates just tested exhaust the relevant contractions. Finding their divergence-free combination still requires a calculation. The principles are simple, but managing the indices and signs is the labor. ^a

^aLecture timestamp: 01:18:42–01:21:44.

1.8 The Field Equation and Its Newtonian Limit

The field equation is now

$$\boxed{R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 8\pi G T_{\mu\nu}}. \quad (1.34)$$

The left-hand side is geometric; the right-hand side describes matter, radiation, and their flows. Their covariant divergences vanish together.

The factor $8\pi G$ is not determined by the divergence argument. It is determined by taking a weak, slowly varying gravitational field sourced by slowly moving matter and requiring the time-time component to reproduce Poisson's equation. In that regime $T_{00} \simeq \rho$, while the other components are suppressed by powers of v/c . The geometric side reduces to the second-derivative structure anticipated in Equation (1.7). The match gives precisely the normalization in Equation (1.34).

This limit also explains why Newtonian gravity appears to be sourced only by mass density. In the full theory, every component of $T_{\mu\nu}$ gravitates: energy density, energy flow, momentum density, pressure, and shear. For ordinary nonrelativistic matter, rest energy dominates. In relativistic matter, radiation, or systems with large pressure, the remaining components cannot be ignored.

The historical route was not a single effortless guess. The Ricci tensor looks like the obvious first attempt, and the extra $-\frac{1}{2}g_{\mu\nu}R$ is what restores compatibility with local conservation. Once the tensor structure is fixed, the Newtonian limit supplies the normalization.

1.9 What Contraction Leaves Behind

Question from the audience. When four indices of the Riemann tensor are contracted to two, does the discarded information have physical content?

Response. Yes. The Ricci tensor does not contain all of spacetime curvature. Source-free gravitational fields provide the clearest examples: the Ricci tensor can vanish while tidal curvature and gravitational waves remain. The contraction keeps the part tied directly to the local source equation, not every degree of freedom of the gravitational field. ^a

^aLecture timestamp: 01:21:57–01:23:10.

The analogy with electromagnetism is useful. Maxwell's equations admit waves in a region containing no charge or current. Likewise, Einstein's equation admits nontrivial geometry in a region where $T_{\mu\nu} = 0$. We now derive the simpler equation that holds there.

1.10 Vacuum Means Ricci-Flat, Not Flat

Set

$$T_{\mu\nu} = 0. \quad (1.35)$$

Then Equation (1.34) gives

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 0. \quad (1.36)$$

Contract with $g^{\mu\nu}$. Since

$$g^{\mu\nu}g_{\mu\nu} = \delta^\mu{}_\mu = 4 \quad (1.37)$$

in four spacetime dimensions, the trace is

$$R - \frac{1}{2}(4)R = -R = 0. \quad (1.38)$$

Therefore

$$R = 0, \quad R_{\mu\nu} = 0. \quad (1.39)$$

The vacuum field equation is Ricci-flatness.

It is essential not to replace the words *Ricci-flat* by *flat*. The Schwarzschild exterior contains no matter away from its central singularity, and it satisfies $R_{\mu\nu} = 0$. Yet its Riemann tensor is nonzero, and tidal forces are real. One can verify Ricci-flatness by calculating all the Christoffel symbols and curvature components by hand, though the calculation is a nuisance, or by letting a symbolic algebra system do the bookkeeping. At the singularity the classical expressions diverge; everywhere away from it, the vacuum equation holds.

Gravitational waves give a second example. They can propagate through a source-free region while carrying genuine spacetime distortion. Thus

$$R_{\mu\nu} = 0 \quad \not\Rightarrow \quad R^\rho{}_{\sigma\mu\nu} = 0. \quad (1.40)$$

1.11 Keeping the Source: The Trace Equation

The lecture nearly ends at the vacuum result, but an audience member asks to repeat the contraction without discarding the source.

Question from the audience. What follows if we perform the same contraction while leaving $T_{\mu\nu}$ in the equation?

Response. The trace relates the scalar curvature directly to the trace of the energy-momentum tensor, and substituting that relation back gives an equivalent Ricci-tensor form of the field equation.^a

^aLecture timestamp: 01:29:28–01:32:20.

Define

$$T = g^{\mu\nu} T_{\mu\nu}. \quad (1.41)$$

Taking the trace of Equation (1.34) gives

$$R - \frac{1}{2}(4)R = 8\pi G T, \quad (1.42)$$

$$-R = 8\pi G T, \quad (1.43)$$

so

$$\boxed{R = -8\pi G T}. \quad (1.44)$$

Substitution back into the field equation yields

$$\boxed{R_{\mu\nu} = 8\pi G \left(T_{\mu\nu} - \frac{1}{2} g_{\mu\nu} T \right)}. \quad (1.45)$$

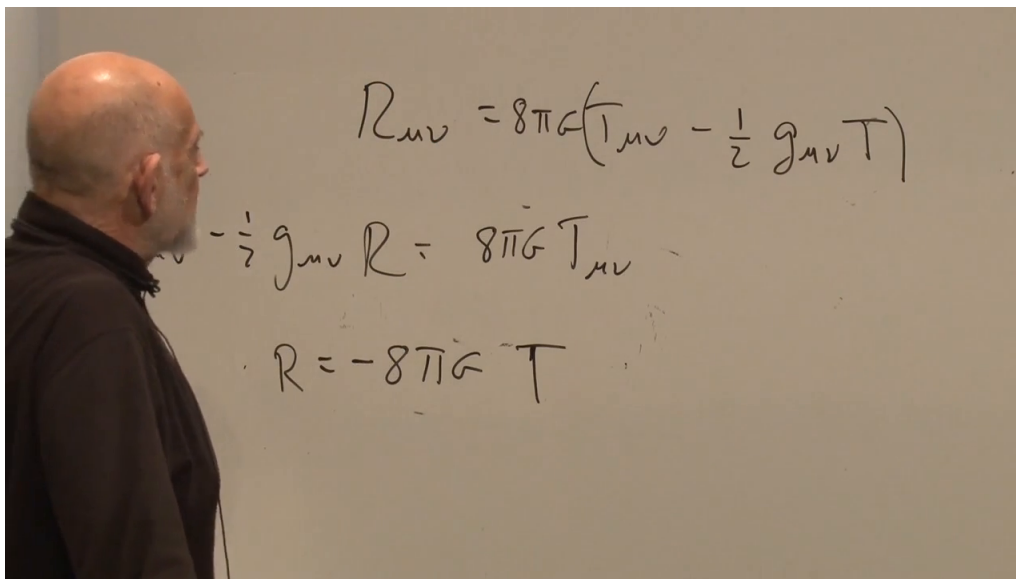


Figure 1.4: The trace relation and the equivalent Ricci-tensor form of Einstein's equation.

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Question from the audience. Does the scalar T have a physical significance?

Response. It is the trace of the energy-momentum tensor. For electromagnetic radiation it vanishes, and the same is true for an ideal gas of massless particles such as photons. Massive matter generally has a nonzero trace. Through Equation (1.44), that trace fixes the scalar curvature. ^a

^aLecture timestamp: 01:32:20–01:33:12.

1.12 How Much Curvature Does Ricci Remember?

Question from the audience. Is there a simple physical or geometric picture of the Ricci tensor and Ricci scalar, comparable to parallel transport around a small loop for the Riemann tensor?

Response. No comparably transparent picture is offered here. The Riemann tensor directly records the rotation produced by transport around a small loop. Its contractions are simpler algebraically, but their individual geometric meanings are harder to visualize. It is better not to invent a misleading picture merely because one would be convenient. ^a

^aLecture timestamp: 01:33:14–01:34:13.

The amount of information lost in the contraction depends on dimension. In four dimensions, the Riemann tensor contains curvature information not present in the Ricci tensor. In three dimensions, the Riemann tensor is determined by the Ricci tensor. In two dimensions, the scalar curvature contains all the independent curvature information. Four-dimensional gravity therefore has room for source-free curvature degrees of freedom that are invisible to $R_{\mu\nu}$.

Question from the audience. If different Riemann tensors can have the same Ricci tensor, how does that missing information appear physically?

Response. A given source does not by itself specify every possible gravitational configuration. Boundary and initial data can include gravitational radiation. In the weak-field approximation one may think of starting with a sourced solution and adding a free gravitational wave. The full nonlinear theory does not permit arbitrary superposition, but the example correctly identifies the missing information: propagating curvature not fixed locally by $T_{\mu\nu}$.^a

^aLecture timestamp: 01:37:05–01:38:59.

The vacuum examples now fit together. Schwarzschild curvature records the tidal field outside a mass; gravitational waves record propagating free curvature. Both can satisfy $R_{\mu\nu} = 0$ without being flat.

1.13 The Cosmological Constant

Question from the audience. Is there no additional fudge factor, such as a cosmological constant?

Response. A cosmological constant is allowed. It can be written on the geometric side as a constant times $g_{\mu\nu}$, or shifted to the right and interpreted as a vacuum contribution to the energy-momentum tensor. The two descriptions are the same equation with different bookkeeping.^a

^aLecture timestamp: 01:39:01–01:40:44.

With a cosmological constant, one common convention is

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \quad (1.46)$$

Moving the extra term to the right defines an effective vacuum stress tensor,

$$T_{\mu\nu}^{(\Lambda)} = -\frac{\Lambda}{8\pi G} g_{\mu\nu}. \quad (1.47)$$

Because metric compatibility gives $D_\mu g^{\mu\nu} = 0$, the extra term is automatically compatible with the conservation law. This is the metric term left open by the earlier uniqueness statement.

1.14 Mercury and the Weak-Field Calculation

Question from the audience. Did Einstein use this equation to calculate Mercury's orbit, and where can one see how that calculation was done?

Response. Einstein did not yet have Schwarzschild's exact solution, but he had a sufficiently accurate large-distance approximation to the exterior field of the Sun. The Sun's radius and Mercury's orbit are enormous compared with the Sun's Schwarzschild radius, so relativistic corrections are small. He could begin with a Keplerian orbit and compute the correction perturbatively. Schwarzschild obtained the exact exterior solution soon afterward, making the same calculation systematic.^a

^aLecture timestamp: 01:40:46–01:43:48.

Perturbation theory means exactly this: start with a problem already solved, then calculate the effect of a small correction. Newton's inverse-square force produces closed Keplerian ellipses. General relativity adds a small departure from the Newtonian radial equation, so the ellipse does not close after one revolution; its perihelion advances.

The lecture closes with the curious specialness of closed central-force orbits. The inverse-square force is exceptional in making the Keplerian orbits close without precession. More generally, Bertrand's theorem also identifies the isotropic harmonic oscillator as the other central potential for which every bound orbit is closed. The small relativistic correction removes that Newtonian degeneracy, turning Mercury's precession into an observable signature of the field equation derived in this chapter.